

Construction of 1,3,5-Triazine-Based Prisms and Their Enhanced Solid-State Emissions

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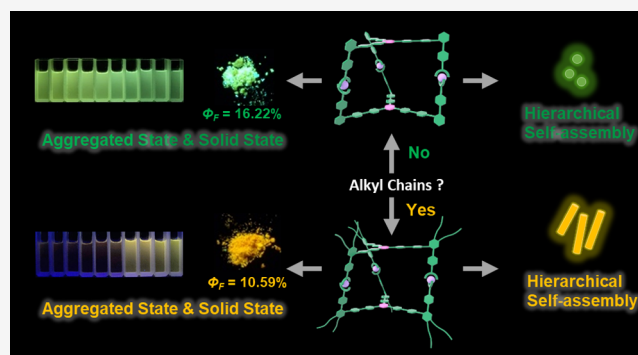


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Supporting Information

ABSTRACT: In this study, two trigonal prisms based on the 1,3,5-triazine motif (SA and SB), distinguished by hydrophobic groups, were prepared by the self-assembly of tritopic terpyridine ligands and Zn(II) ions. SA and SB exhibited high luminescence efficiencies in the solid state, overcoming the fluorescence quenching of the 1,3,5-triazine group caused by π – π interactions. Notably, SA and SB exhibited different luminescence behaviors in the solution state and aggregation state. SB with 12 alkyl chains exhibited extremely weak fluorescence in a dilute solution, but its fluorescence intensity and photoluminescence quantum yield (PLQY) were significantly enhanced in the aggregated state (with the increase in the water fraction), especially in the solid state. Different from the gradually enhanced efficiency of SB, the PLQY of SA gradually decreased with the increase in aggregation but still maintained a high luminescence efficiency. These two complexes exhibited different modes to solve the fluorescence quenching of 1,3,5-triazine in the solid state. The hierarchical self-assembly of SB exhibited nanorods owing to the hydrophobic interactions of alkyl chains, while SA aggregated into spheres under the influence of π – π interactions.



INTRODUCTION

In nature, the coordination of metal ions with organic substrates has been exploited to create various functional compounds, such as chlorophyll,¹ vitamin B₁₂,² hemoglobin,³ etc. Compared to other weak noncovalent interactions, coordination bonds are advantageous in terms of their directionality and predictability.^{4–6} Inspired by nature, chemists have utilized coordination-driven self-assembly strategies to construct discrete architectures from two-dimensional (2D) rings to three-dimensional (3D) cages with precise sizes and shapes,^{7–17} which have applications in areas such as catalysis,¹⁸ cancer treatment,¹⁹ and optical materials;^{20,21} among these architectures, 2,2':6',2''-terpyridine (tpy) as a building block has gained widespread attention owing to its remarkable binding ability to different transition-metal ions.^{17,22–24} Compared to 2D rings, 3D nanocages demonstrate promising applications for anion recognition,^{25–28} C₆₀ encapsulation,^{29–33} and controlled release^{34–37} because of their regular configuration with fixed-size cavities that can selectively encapsulate guest molecules.³⁸

In recent years, luminescent materials have considerably facilitated and enriched human lives, and the preparation of efficient solid-state luminescent materials has been the focus of scientific research. The stacking pattern of molecules is one of the most important factors affecting the optical behavior of

molecules. The strategies, including cocrystal,³⁹ host–guest interaction,⁴⁰ and steric hindrance,⁴¹ were employed to enhance solid-state emission by controlling of molecular packing.⁴² In this regard, 1,3,5-triazine motifs with good thermal stability and electrophilic properties^{43–45} are often used as acceptor units in phosphor emitters,^{45–47} organic light-emitting diodes (OLED),^{48,49} and fluorescence sensing.^{50–52} However, owing to the high planarity of the triazine ring, fluorescence is severely quenched in the solid state because of strong π – π stacking interactions, which considerably affects the luminescent properties of the materials. Therefore, it is imperative to develop a strategy to avoid the dense packing of triazine rings and achieve efficient solid-state luminescence.

Herein, two 1,3,5-triazine-based supramolecular trigonal prisms SA and SB were constructed by tritopic ligands LA (without alkyl chains) and LB (with six alkyl chains at the corners) coordinated with Zn(II) in a precise ratio of 2:3, respectively. In addition, their photophysical properties in a

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dilute solution, in the aggregated state (with the increase in the water fraction), and in the solid state were investigated. Notably, these two trigonal prisms exhibited considerably different luminescence efficiencies in the dilute solution, while they exhibited high photoluminescence quantum yields (PLQYs) in the solid state. In addition, to clarify the effect of hydrophobic interactions, the hierarchical assembly behaviors of the trigonal prisms were further investigated. The transmission electron microscopy (TEM) images showed that **SA** aggregated into nanospheres, while **SB** with alkyl chains formed nanorods induced by the interactions of hydrophobic chains.

EXPERIMENTAL SECTION

Compound 2 and compound 3: The synthetic procedure was reported in ref 53. Nuclear magnetic resonance (NMR) spectra data were recorded at 300 K on Bruker 500 MHz, Qone AS 400 MHz nuclear magnetic resonance instruments using CDCl₃ and CD₃CN as the solvents. Electrospray ionization mass spectrometry (ESI-MS) spectra were collected on a Waters Synapt G2 mass spectrometer, using solutions of 0.5 mg sample in 1 mL of acetonitrile for the complexes. MALDI-TOF-MS was performed on a Bruker AutoflexIII using trans-2-[3-(4-*tert*-butylphenyl)-2-methyl-2-propenyldene]-malononitrile (DCTB) as a matrix. The matrix dissolved in CHCl₃ at 20 mg mL⁻¹ and organic compounds were dissolved in CHCl₃ at 10 mg mL⁻¹. The sample was prepared by depositing 0.5 μ L of matrix on the wells of a 384-well ground-steel plate, allowing the spots to dry, depositing 0.5 μ L of the sample on a spot of dry matrix, and adding another 0.5 μ L of matrix on top of the dry sample. The plate was inserted into the MALDI source after drying. The sample was conducted in linear mode. And the data analysis was 23 conducted with Bruker's FlexAnalysis software. The method for the preparation of monodisperse samples: The sample was dissolved in CH₃CN at concentrations of 5×10^{-6} M. The solutions were drop-cast onto a copper grid (ultrathin carbon supported by a lacey carbon film on a 200 mesh copper grid), and extra solution was absorbed by filter paper to avoid aggregation. The method for the preparation of hierarchical assembly samples: 2.7 mL of water was slowly added to 0.3 mL of acetonitrile solution by the peristaltic pump at a flow rate of 15 μ L/min. Then, 10 μ L of sample solution was dropped on the copper grids to prepare samples. The TEM images of the drop-cast samples were taken with a JEM-2100F transmission electron microscope. Atomic force microscope microscopy (AFM) imaging was performed on a Bruker Dimension Icon AFM system with ScanAsyst, under tapping mode and the type of probe is RTESPA-300. The data were processed by NanoScope Analysis version 2.0 (Bruker Software, Inc.). The sample was dissolved in CH₃CN at a concentration of 10^{-6} M and then dropped the diluted solution onto a fresh mica wafer. All ultraviolet–visible (UV–vis) spectra of solution samples were recorded on a PerkinElmer Lambda 365 spectrophotometer at room temperature. All emission spectra of solution samples were recorded on a Shimadzu RF-S301pc spectrophotometer. All emission spectra of solid samples and PLQYs were determined by an FLS980 spectrometer (Edinburgh Instruments Ltd., Livingston, United Kingdom). PLQY was measured via an integrating light sphere. Solutions were placed in 1 cm path length quartz cells.

Synthesis of Ligand LA. A mixture of 2,4,6-tris(4-bromophenyl)-1,3,5-triazine (50 mg, 0.03 mmol), compound 3 (70.5 mg, 0.12 mmol), and Pd(PPh₃)₂Cl₂ (3.5 mg, 0.005 mmol) was added in 100 mL Schlenk flask. Toluene (20 mL), 1 M Na₂CO₃ aqueous solution (10 mL), and *tert*-butyl alcohol (3 mL) were added under N₂. The mixture was kept at 85 °C for 3 days. After cooling to room temperature, the solution was extracted by CHCl₃ three times and the solvent was removed under reduced pressure. The crude product was purified by column chromatography on silica gel (CH₂Cl₂/EtOH = 40/1) to give ligand **LA** as a white solid (23.3 mg, 46%). The crude product was purified by column. ¹H NMR (400 MHz, Chloroform-*d*) δ 8.81 (d, *J* = 8.1 Hz, 6H, Ph-*H*^a), 8.74 (s, 6H, tpy-*H*^{3',5'}), 8.69 (dd, *J*

= 4.8, 1.6 Hz, 6H, tpy-*H*^{6,6''}), 8.65 (d, *J* = 8.0 Hz, 6H, tpy-*H*^{3,3''}), 7.87–7.81 (m, 18H, tpy-*H*^{4,4''}, Ph-*H*ⁱ, and Ph-*H*^j), 7.62 (d, *J* = 8.0 Hz, 6H, Ph-*H*^b), 7.54–7.47 (m, 12H, Ph-*H*^c, Ph-*H*^e, and Ph-*H*^f), 7.36 (d, *J* = 8.0 Hz, 6H, Ph-*H*^b), 7.31 (dd, *J* = 7.7, 4.1 Hz, 12H, tpy-*H*^{5,5''}, and Ph-*H*^g). ¹³C NMR (100 MHz, Chloroform-*d*) δ 171.32, 156.25, 155.90, 149.88, 149.13, 144.55, 142.40, 141.18, 140.10, 139.96, 138.49, 136.95, 136.56, 135.17, 130.81, 130.51, 130.48, 129.47, 127.92, 127.82, 127.21, 127.07, 126.95, 123.88, 121.39, 118.81. MALDI-TOF: calcd for [C₁₂₀H₇₈N₁₂+H]⁺ 1687.65 Da (found 1687.66 Da).

Synthesis of Ligand LB. A mixture of 2,4,6-tris(4-bromophenyl)-1,3,5-triazine (50 mg, 0.03 mmol), compound 2 (94.5 mg, 0.12 mmol), and Pd(PPh₃)₂Cl₂ (3.5 mg, 0.005 mmol) was added in 100 mL Schlenk flask. Toluene (20 mL), 1 M Na₂CO₃ aqueous solution (10 mL), and *tert*-butyl alcohol (3 mL) were added under N₂. The mixture was kept at 85 °C for 3 days. After cooling to room temperature, the solution was extracted by CHCl₃ three times and the solvent was removed under reduced pressure. The crude product was purified by column chromatography on silica gel (CH₂Cl₂/EtOH = 50/1) to give ligand **LB** as a white solid (35.0 mg, 51%). The crude product was purified by column. ¹H NMR (400 MHz, Chloroform-*d*) δ 8.81 (d, *J* = 8.1 Hz, 6H, Ph-*H*^b), 8.74 (s, 6H, tpy-*H*^{3',5'}), 8.70 (d, *J* = 4.9 Hz, 6H, tpy-*H*^{6,6''}), 8.65 (d, *J* = 8.0 Hz, 6H, tpy-*H*^{3,3''}), 7.88–7.80 (m, 18H, tpy-*H*^{4,4''}, Ph-*H*^g, and Ph-*H*^j), 7.61 (d, *J* = 8.0 Hz, 6H, Ph-*H*^f), 7.36–7.28 (m, 18H, tpy-*H*^{5,5''}, Ph-*H*^b, and Ph-*H*^e), 7.03 (d, *J* = 1.7 Hz, 6H, Ph-*H*^c, and Ph-*H*^d), 4.12 (q, *J* = 6.1 Hz, 12H, Alkyl-*H*ⁱ, and Alkyl-*H*^j), 1.92–1.85 (m, 12H, Alkyl-*H*ⁱ, and Alkyl-*H*^j), 1.53 (s, 12H, Alkyl-*H*^k, and Alkyl-*H*^k), 1.41–1.34 (m, 24H, Alkyl-*H*ⁱ, Alkyl-*H*^m, Alkyl-*H*^l, and Alkyl-*H*^m), 0.95–0.90 (m, 18H, Alkyl-*H*ⁿ, and Alkyl-*H*^m). ¹³C NMR (100 MHz, Chloroform-*d*) δ 171.34, 156.28, 155.91, 149.93, 149.15, 148.75, 148.69, 144.63, 142.48, 141.29, 138.18, 136.94, 136.24, 135.13, 132.66, 132.49, 130.59, 130.56, 129.48, 127.20, 127.03, 126.93, 123.87, 121.39, 118.77, 116.22, 116.17, 69.59, 69.55, 31.68, 29.38, 25.82, 22.72, 14.14. MALDI-TOF: calcd for [C₁₅₆H₁₅₀N₁₂O₆]⁺ 2287.18 Da (found 2287.19 Da).

Synthesis of Complex SA. To a solution of ligand **LA** (6.0 mg, 3.56 μ mol) in CHCl₃ (3 mL), a solution of Zn(NO₃)₂·6H₂O (1.6 mg, 5.34 μ mol) in MeOH (3 mL) was added, and then the mixture was stirred at 50 °C for 12 h. After cooling to room temperature, excess NH₄PF₆ (44 mg) was added to generate an orange precipitate (7.3 mg, 92%). ¹H NMR (400 MHz, Acetonitrile-*d*₃) δ 8.97 (s, 6H, tpy-*H*^{3',5'}), 8.87 (d, *J* = 8.2 Hz, 6H, Ph-*H*^b), 8.69 (d, *J* = 8.3 Hz, 6H, tpy-*H*^{3,3''}), 8.17–8.10 (m, 12H, tpy-*H*^{4,4''} and Ph-*H*^a), 8.00 (d, *J* = 8.2 Hz, 6H, Ph-*H*^f), 7.81 (t, *J* = 8.0 Hz, 12H, tpy-*H*^{6,6''}, and Ph-*H*^b), 7.63 (d, *J* = 6.8 Hz, 18H, Ph-*H*^{c,d,e,f,b}), 7.48 (d, *J* = 7.9 Hz, 6H, Ph-*H*^g), 7.36 (t, *J* = 6.3 Hz, 6H, tpy-*H*^{5,5''}). ESI-MS (*m/z*): 1336.7 Da [M-3PF₆]³⁺ (calcd *m/z*: 1336.7 Da), 966.5 Da [M-4PF₆]⁴⁺ (calcd *m/z*: 966.5 Da), 744.4 Da [M-5PF₆]⁵⁺ (calcd *m/z*: 744.4 Da), 596.4 Da [M-6PF₆]⁶⁺ (calcd *m/z*: 596.4 Da).

Synthesis of Complex SB. To a solution of ligand **LB** (6.0 mg, 2.62 μ mol) in CHCl₃ (3 mL), a solution of Zn(NO₃)₂·6H₂O (1.2 mg, 3.93 μ mol) in MeOH (3 mL) was added, and then the mixture was stirred at 50 °C for 12 h. After cooling to room temperature, excess NH₄PF₆ (32 mg) was added to generate an orange precipitate (6.6 mg, 89%). ¹H NMR (400 MHz, Acetonitrile-*d*₃) δ 8.96 (s, 12H, tpy-*H*^{3',5'}), 8.87 (d, *J* = 8.2 Hz, 12H, Ph-*H*^b), 8.69 (d, *J* = 8.2 Hz, 12H, tpy-*H*^{3,3''}), 8.13 (t, *J* = 8.5 Hz, 24H, tpy-*H*^{4,4''} and Ph-*H*^a), 8.00 (d, *J* = 8.3 Hz, 12H, Ph-*H*^g), 7.80 (d, *J* = 6.1 Hz, 24H, tpy-*H*^{6,6''}, and Ph-*H*^f), 7.62 (d, *J* = 7.8 Hz, 12H, Ph-*H*^b), 7.45 (d, *J* = 8.0 Hz, 12H, Ph-*H*^e), 7.36 (s, 12H, tpy-*H*^{5,5''}), 7.14 (d, *J* = 15.1 Hz, 12H, Ph-*H*^c, and Ph-*H*^d), 4.18 (s, 24H, Alkyl-*H*ⁱ, and Alkyl-*H*^j), 1.87 (d, *J* = 7.3 Hz, 24H, Alkyl-*H*ⁱ, and Alkyl-*H*^j), 1.57 (s, 24H, Alkyl-*H*^k, and Alkyl-*H*^k), 1.43 (s, 48H Alkyl-*H*ⁱ, Alkyl-*H*^m, Alkyl-*H*^l, and Alkyl-*H*^m), 0.97 (d, *J* = 5.9 Hz, 35H, Alkyl-*H*ⁿ, and Alkyl-*H*ⁿ). ESI-MS (*m/z*): 1737.0 Da [M-3PF₆]³⁺ (calcd *m/z*: 1737.0 Da), 1266.8 Da [M-4PF₆]⁴⁺ (calcd *m/z*: 1266.8 Da), 984.6 Da [M-5PF₆]⁵⁺ (calcd *m/z*: 984.6 Da).

Scheme 1. Self-Assembly of Trigonal Prisms SA and SB.

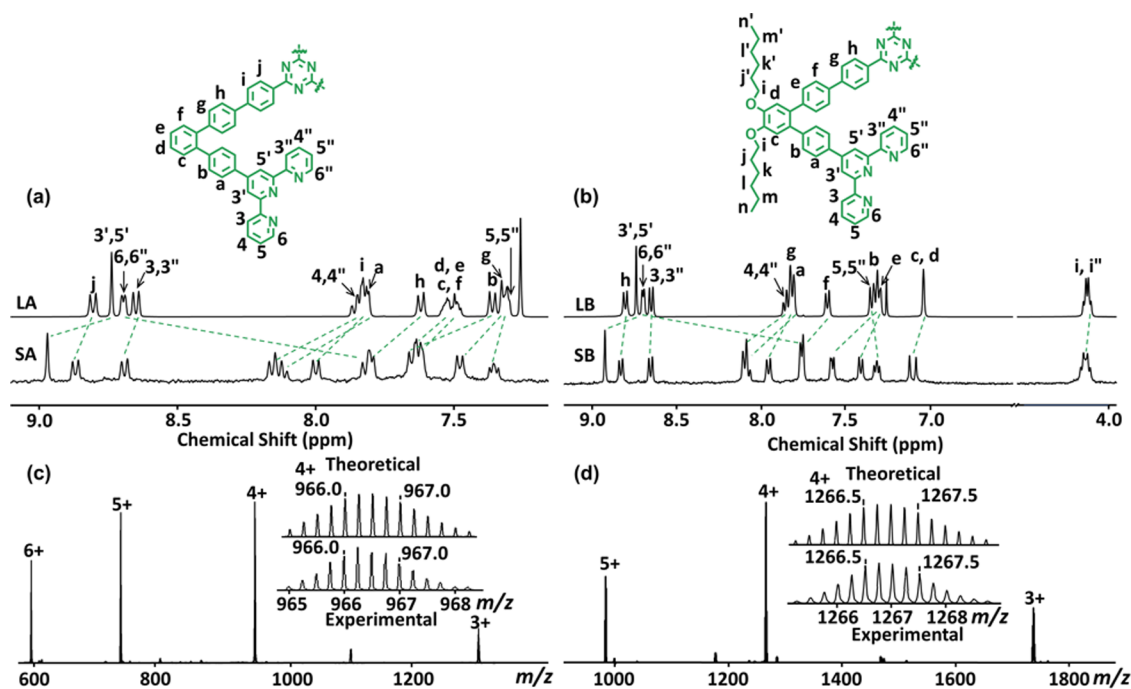
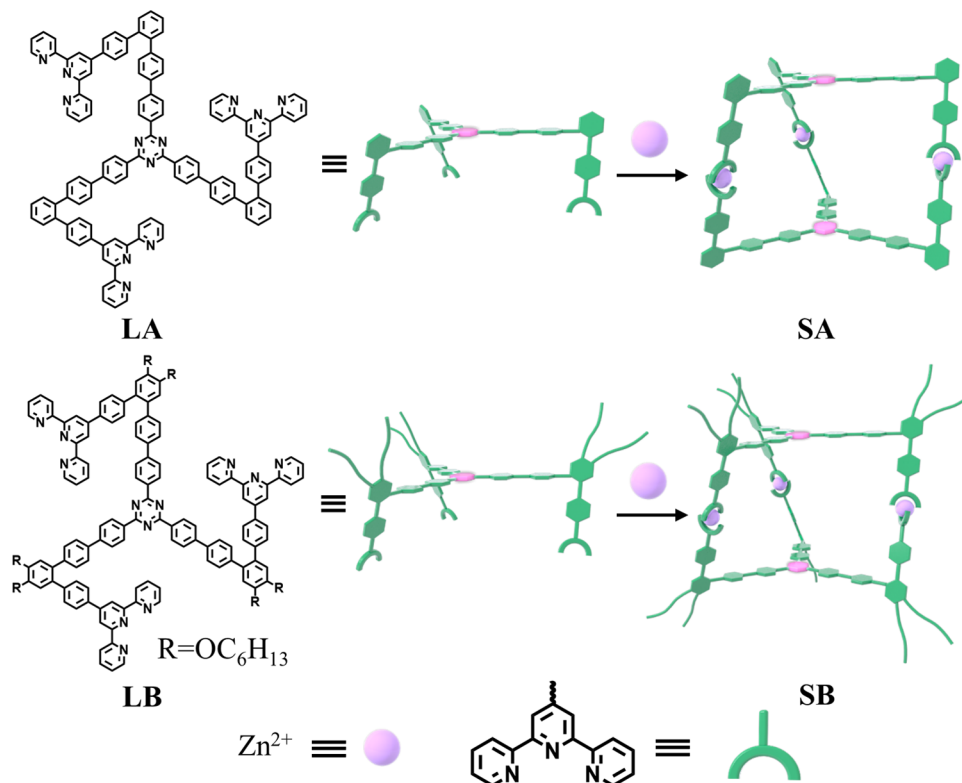


Figure 1. ^1H NMR spectra (400 MHz, 300 K) of (a) LA in CDCl_3 and SA in CD_3CN and (b) LB in CDCl_3 and SB in CD_3CN . ESI-MS spectra of (c) SA and (d) SB.

RESULT AND DISCUSSION

LA was obtained by the Suzuki coupling reaction of intermediate **2**⁵³ (Supporting Information Scheme S1) and 2,4,6-tris(4-bromophenyl)-1,3,5-triazine and purified by column chromatography (SiO_2) to obtain the final product in 46% yield. To further investigate the hydrophobic effect on the

luminescence behaviors and hierarchical assembly, six C6 alkyl chains were introduced at the corners of as-obtained LB (Supporting information Scheme S1, yield: 51%) by the same synthetic method to provide sites for hydrophobic interactions. LA and LB were assembled with $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ in a precise ratio of 2:3 in a mixed solvent of $\text{CHCl}_3/\text{MeOH}$ (1:1, v/v) at 50 °C for 12 h, respectively. Then, excess NH_4PF_6 was added,

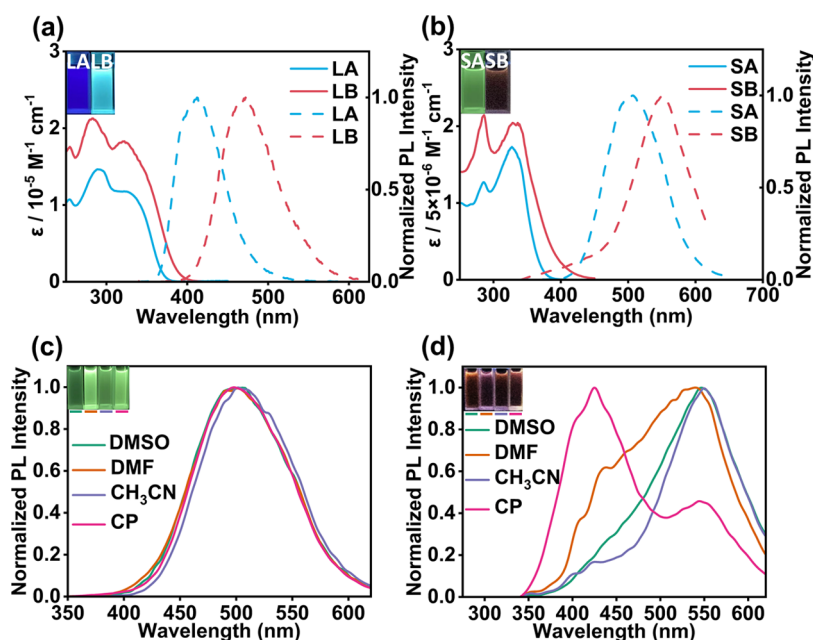


Figure 2. (a) UV-vis and PL spectra of LA and LB in CHCl_3 ($\lambda_{\text{ex}} = 320$ nm, $c = 10^{-5}$ M). (b) UV-vis and PL spectra of SA and SB in acetonitrile ($\lambda_{\text{ex}} = 320$ nm, $c = 5 \times 10^{-6}$ M). Solvent effect on the light-emitting properties of (c) SA and (d) SB ($\lambda_{\text{ex}} = 320$ nm, $c = 5 \times 10^{-6}$ M). DMSO, dimethyl sulfoxide; DMF, *N,N*-dimethylformamide; CH_3CN , acetonitrile; CP, acetone.

affording a yellow precipitate in high yields of 92% for SA and 89% for SB (Scheme 1).

In ^1H NMR spectra of all ligands and complexes, only one set of signals was observed for tpy, indicative of the formation of highly symmetric molecules (Figure 1a,b). The proton signals of the ligands and obtained complexes were further confirmed by 2D COSY (Figures S3, S8, S10, S13), and the proton signals of tpy- $\text{H}^{3',5'}$ in SA and SB were shifted downfield ($\delta \approx 0.24$ ppm) because of the lower electron density compared with those of their ligands.⁵⁴ However, owing to the electron shielding effect, the peaks of tpy- $\text{H}^{6,6''}$ shifted upfield significantly as expected ($\delta \approx 0.90$ ppm).⁵⁵ These changes in the chemical shift of these signals indicated the coordination between tpy units and Zn(II) ions.

The compositions of SA and SB were further characterized by electrospray ionization mass spectrometry (ESI-MS). Owing to the successive loss of different numbers of PF_6^- , only one prominent set of peaks with different charge states was observed in the ESI-MS spectra of SA and SB (Figure 1c,d). After deconvolution, the molecular weights of two triprisms SA and SB were 4444.8 and 5646.4 Da, respectively, which were in good agreement with the expected formation of two ligands, three Zn(II) ions, and six PF_6^- ions. The isotope patterns of each peak matched well with their corresponding simulated isotope patterns (Figures S15 and S16).

To further confirm the formation of SA and SB, the sizes of the trigonal prisms were visually characterized by transmission electron microscopy (TEM). The acetonitrile solutions of complexes SA and SB were dropped onto carbon-coated grids, followed by insertion of the dried sample into a specimen bin under vacuum. The circular profiles of single molecules with diameters of 2.9 nm (SA) on the carbon films were observed, which were in perfect agreement with the diameters of 2.9 nm for the optimized prism molecule models. Owing to the presence of alkyl chains, the observed diameter of SB (3.1 nm) was slightly greater than that of SA, but still within a reasonable theoretical diameter range (Figure S17). Atomic force

microscope microscopy (AFM) images of SA and SB further provided evidence for the formation of SA and SB on a newly cut mica surface. From the results, the average heights of SA and SB were 2.8 and 2.9 nm, respectively, further supporting the formation of discrete structures (Figure S18).

After confirming the composition of the complexes, the photophysical properties of the ligands and supramolecules were further investigated. As shown in the absorption spectra (Figure 2a,b) of LA and LB in chloroform ($c = 10^{-5}$ M), two primary peaks at 285 and 320 nm were attributed to tpy and 1,3,5-triazine motifs. After coordination with Zn(II) ions, the absorption peaks of SA and SB ($c = 5 \times 10^{-6}$ M) at 320 nm were slightly red-shifted to 330 nm, accompanied by peak enhancement. The change in absorption wavelength may be caused by the increase in solvent polarity and the change in absorption peak profile may be attributed to the change in ligand-ligand interactions after coordination. Compared with the fluorescence emission peak of ligand LA at 410 nm, the emission peak of LB (480 nm) was significantly red-shifted due to intramolecular charge transfer (ICT). The prism SA exhibited a single emission with an emission band at 500 nm, and the main emission band of SB was observed at 550 nm, which was attributed to ICT. SA exhibited a strong emission in acetonitrile with an absolute quantum yield (Φ_F) of 39.18%, which was significantly greater than that of SB (0.07%). Therefore, the introduction of C6 alkyl chains is detrimental to the luminescence of the complexes in the solution state.

Furthermore, the fluorescence properties of complexes SA and SB were investigated in different polar solvents. With the variation in solvent polarity, SA and SB exhibited different fluorescence responses. After normalization, SA showed basically overlapped fluorescence spectra in different solvents and lime green luminescence (Figure 2c). However, the emission peak of SB changed, which is owing to solvents with different polarities having effects on ICT (Figure 2d), further

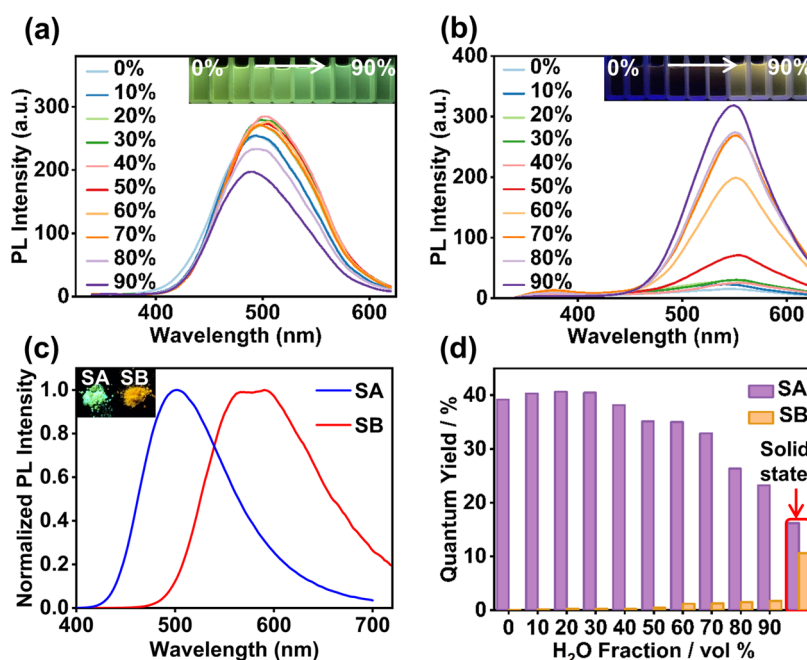


Figure 3. PL spectra of (a) SA and (b) SB versus H₂O in CH₃CN/H₂O mixtures ($\lambda_{\text{ex}} = 320$ nm, $c = 5 \times 10^{-6}$ M); normalized PL spectra of (c) SA and SB in the solid state ($\lambda_{\text{ex}} = 320$ nm) with inset photographs of SA and SB in the solid state under excitation using a 365 nm UV lamp at 300 K; (d) quantum yields of SA and SB at 320 nm with different fractions of water in the CH₃CN/H₂O solution.

having influence on the ratios of LE and ICT state emission, and ultimately changing the fluorescence color.^{56,57}

To compare the effect of hydrophobic chains of trigonal prisms on the optical behaviors in the aggregated state, fluorescence emission spectra were recorded in a mixed solution of acetonitrile as the good solvent and water as the poor solvent (Figure 3a,b). At a water fraction of 20%, the fluorescence intensity of SA increased slightly, and with the increase in the water fraction, the fluorescence intensity was basically unchanged. At the water fraction up to 80%, the fluorescence intensity decreased significantly owing to the formation of tightly packed agglomerates. SB exhibited different fluorescence behavior with the increase of water fractions. The fluorescence intensity increased until it reached a maximum at the water fraction of 90%, with yellow emission and a Φ_F of 1.72%; this value was still considerably less than that of SA ($\Phi_F = 23.23\%$) under the same conditions (Figure 3d). The emission wavelengths of neither SA nor SB shifted significantly during this process.

Apart from the optical properties in the solution, the optical properties of the ligands and complexes in the solid state were investigated. LA exhibited a red shift of ca. 20 nm in the fluorescence spectrum, whereas the fluorescent band of LB was unchanged (Figure S20). As for the complexes, the fluorescent band of SA was unchanged, and the excessive aggregation of molecules caused a significant decrease in the PLQY compared with that observed in the dilute solution, but it still maintained a high Φ_F value ($\Phi_F = 16.22\%$). SB exhibited a broad peak at 560–600 nm and a significant increase in the PLQY ($\Phi_F = 10.59\%$) in comparison with that observed in the mixed solvent system of acetonitrile and water (Figure 3c,d). Benefitting from the limitation of the prismatic structure and further hindrance of molecular aggregation by the alkyl chains are thought to be responsible for hindering tight stacking of the triazine moieties; thus, SA and SB exhibited efficient

luminescence in the solid state compared to the triazine molecules in the solid state.

The variation in the spectra of trigonal prisms in the aggregated state caused by hydrophobic chains was often accompanied by differences in the formation of aggregates via hierarchical self-assembly, which was also verified by TEM images (Figure 4). In a mixed solution of CH₃CN/H₂O (1:9,

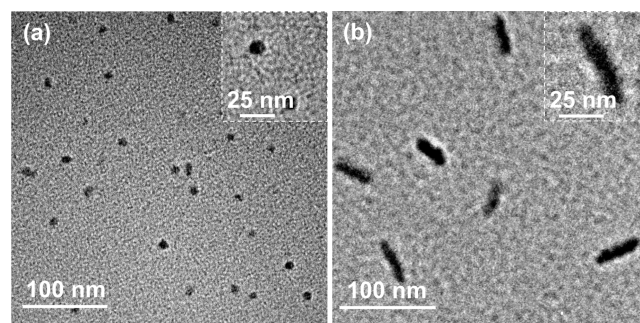


Figure 4. TEM images of hierarchical self-assembly formed by (a) SA and (b) SB.

v/v), SA exhibited uniform spherical aggregates with a diameter of ca. 13.5 nm, while SB formed nanorods with a diameter of ca. 8.5 nm and a length ranging from 35 to 50 nm. The formation of SA aggregates was assumed to be dominated by the π – π interactions between the 1,3,5-triazine and tpy motifs. In the case of SB in the mixed solvent, the C6 chains were entangled and aggregated by hydrophobic interactions, which could reduce the triazine motif packing tightly and maintain high quantum yield in the solid state.

CONCLUSIONS

In summary, two trigonal prisms SA and SB based on the 1,3,5-triazine motifs were successfully constructed, and their

luminescence properties were investigated. In a dilute solution, **SA** exhibited efficient emission, while **SB** with alkyl chains exhibited almost no fluorescence. Meanwhile, in the aggregated state, although the fluorescence intensity of **SA** decreased in the aggregated and solid states, it still exhibited a high PLQY; on the contrary, the fluorescence intensity of **SB** increased continuously with the increase of poor solvents and exhibited the highest PLQY in the solid state. Owing to the prismatic backbone and the restriction of hydrophobic chains, **SA** and **SB** prevented fluorescence quenching and exhibited high PLQYs even in the solid state. Moreover, TEM results of **SA** and **SB** revealed the effect of hydrophobic chains on molecular stacking. **SA** formed spherical aggregates, while **SB** was influenced by the interaction of hydrophobic chains formed nanorods. This study provided completely different insights into the development of luminescent materials.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.inorgchem.3c00420>.

Synthetic and characterization data (NMR, ESI-MS, fluorescence spectra, photoluminescence quantum yields, TEM images, and AFM images) (PDF)

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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